

Processors and memory

Inside a computer are many types of electronic chips. Most importantly, the processor chip runs programs and memory chips store data for instant access.

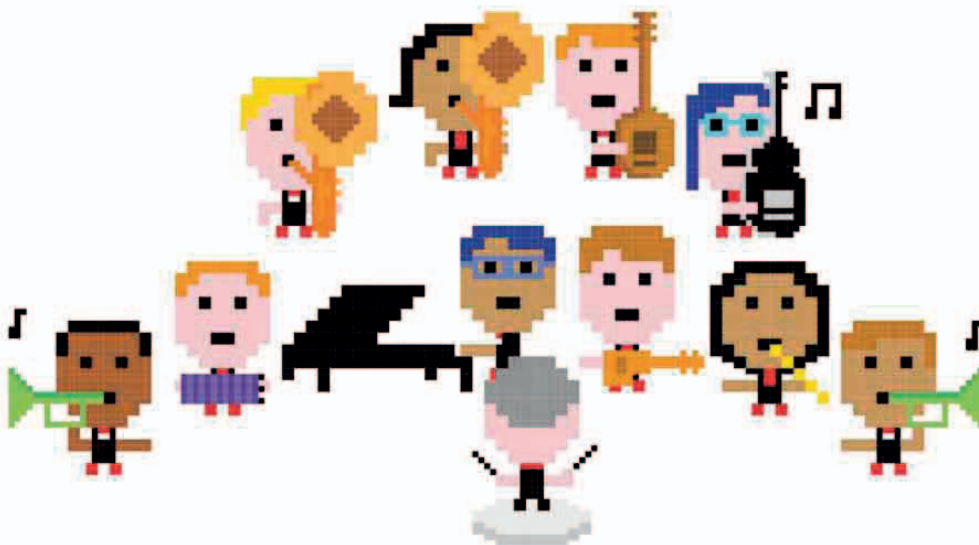
SEE ALSO

◀ **180–181** Inside a computer

◀ **186–187** Logic gates

The processor

Processors are a collection of very small and complex circuits, printed on a glasslike material called silicon. Small switches called transistors are combined to form simple logic gates, which are further combined to form complex circuits. These circuits run all the programs on your computer.



◀ Circuits in a processor

The circuits are kept synchronized by a clock pulse, just like an orchestra is kept in time by a conductor.

Machine code

Processors only understand a set of program instructions called “machine code”. These simple instructions for operations like adding, subtracting, and storing data are combined to create complex programs.

▷ Understanding machine code

Machine code is just numbers, so coders use programming languages like Python that get converted into machine code.

Save to memory

Call another piece of code

Compare two values

